

NDSS

Overall Review

Network and Distributed System Security Symposium: Practical Constraints and Operational Realities (2022-2024)

Research at the Network and Distributed System Security (NDSS) Symposium from 2022 through 2024 reveals a critical gap between theoretical adversarial machine learning research and the operational complexities of real-world deployments. Unlike work focusing primarily on model-level vulnerabilities, NDSS contributions emphasize system-level constraints, distributed training environments, and domain-specific physical realities that fundamentally shape both attack feasibility and defense viability.

From Digital Perturbations to Physical Deployment Constraints

A fundamental limitation of conventional adversarial machine learning research lies in its focus on digital adversarial examples that fail to account for the complex physical conditions governing real-world perception systems. While theoretical studies have produced prolific results on digitally crafted perturbations, these approaches often ignore how distance, viewing angle, and illumination changes severely diminish attack effectiveness when adversarial examples are printed and deployed in physical environments. This disconnect is particularly critical for safety-critical applications—autonomous vehicles operating Traffic Sign Recognition systems cannot be tested solely against digital perturbations displayed on screens in controlled laboratory conditions.

Recent research directly addresses this gap through systematic pipelines for generating robust physical adversarial examples capable of withstanding real-world environmental variations. Extensive real-road driving tests against YOLO v5-based Traffic Sign Recognition systems demonstrated successful attacks against production-grade 2021 model vehicles, proving that theoretical vulnerabilities can indeed translate into tangible threats to deployed automotive technology (Jia et al., 2022). Beyond automotive contexts, adversarial research increasingly incorporates domain-specific physical constraints. For video recognition systems deployed in streaming services, attacks must overcome the inherent denoising effect of standard video compression algorithms, which automatically invalidates most digital perturbations designed without accounting for this preprocessing step (Chang et al., 2023). Similarly, 3D adversarial objects targeting X-ray prohibited item detection systems must account for complex object overlap in luggage and the specific imaging principles of X-ray technology (Liu et al., 2023).

Inappropriate Metrics and Evaluation Frameworks

A significant contributor to the research-practice gap stems from academic reliance on simplified or domain-inappropriate evaluation metrics. The conventional focus on mathematical imperceptibility measures—typically L_p norms quantifying pixel-level differences—proves largely irrelevant for real-world applications operating on non-image data. In fraud detection systems processing tabular financial data, for instance, the meaningful attack constraint is not whether perturbations are visually imperceptible but rather the quantifiable financial cost or utility an adversary must expend to execute a successful attack (Kireev et al., 2023). This mismatch extends to robustness certification: traditional methods designed around single metrics like L_p norms cannot accommodate the highly heterogeneous feature spaces and strict real-time operational requirements of industrial systems like deep learning-based traffic analysis (Wang et al., 2023).

Detection systems suffer from similarly inappropriate evaluation paradigms. The rapid adoption of large language models like ChatGPT created urgent societal demand for detection tools addressing concerns like plagiarism and misinformation, yet existing detectors frequently lack both precision and generality when deployed beyond their training distributions (Kumari et al., 2024). More fundamentally, academic evaluations of defenses rely on manually tuned adaptive adversary models that fail to capture the complex, multi-constraint

optimization problems real attackers face. Conventional threat modeling inadequately represents sophisticated adversaries who must simultaneously optimize attack success, minimize detection risk, and satisfy operational constraints—a limitation addressed by automated evaluation frameworks like AutoAdapt that systematically explore realistic constraint spaces (Krauß et al., 2024).

Furthermore, foundational datasets used to develop security mechanisms often conflate adversarial behaviors with benign ones. Research into browser fingerprinting revealed that datasets treat legitimate users with privacy-conscious browsing configurations identically to sophisticated attackers, leading to fundamentally inaccurate conclusions about real-world attack prevalence and defense requirements (Wu et al., 2023). Such methodological flaws undermine the validity of both attack demonstrations and defensive evaluations.

Architectural and Deployment Complexity

Real-world machine learning deployments introduce architectural constraints that academic research frequently overlooks. Modern deep neural networks undergo substantial transformation between training and deployment: models developed in high-level frameworks like TensorFlow or PyTorch are compiled into optimized executables for edge devices and cloud services. Security mechanisms integrated solely at the framework level fail once models are compiled, necessitating defenses embedded directly within the deep learning compiler pipeline to ensure protection persists through deployment (Chen et al., 2023). This architectural challenge is particularly acute for emerging service models like Encoder as a Service (EaaS), where clients lack white-box access to encoder internals and face prohibitive communication overhead for rigorous certified robustness verification, rendering conventional defense techniques infeasible (Qu et al., 2023).

Distributed training environments compound these challenges. Federated Learning deployments exhibit inherent statistical heterogeneity across client data distributions—a reality that breaks traditional defense mechanisms relying on outlier detection. Such defenses falsely exclude benign models from heterogeneous (non-IID) data sources, causing unacceptable utility degradation for legitimate clients with naturally deviating distributions (Rieger et al., 2022). Effective defenses must therefore leverage this heterogeneity rather than treating

it as noise. DeepSight demonstrates this principle by utilizing model inspection to identify malicious updates based on homogeneity artifacts, successfully mitigating sophisticated backdoor attacks without penalizing clients with legitimately different data distributions (Rieger et al., 2022). Similarly, defenses targeting privacy leakage in federated settings must explicitly account for the statistical heterogeneity inherent to these distributed pipelines rather than assuming uniform data distributions (Zhou et al., 2023).

Stealthy Attacks and Evolving Threat Models

The practical deployment environment incentivizes fundamentally different adversarial strategies than those studied in controlled academic settings. In collaborative training environments like Federated Learning, attackers prioritize stealth over attack sophistication to avoid detection during extended training periods. Model Hijacking attacks exemplify this shift: rather than introducing obvious distributional anomalies, attackers employ novel model structures—the Camouflager—to make malicious data samples appear visually and statistically similar to benign training data while maintaining utility for the original task, enabling execution of a completely separate malicious task without detection (Salem et al., 2022).

Traditional defenses fail against such adaptive threats because they rely on static assumptions about attack characteristics. Backdoor defenses assuming universal, sample-agnostic triggers become useless against modern dynamic backdoor attacks that violate these core premises (Ma et al., 2023). Similarly, audio defenses designed against simple noise-based perturbations fundamentally misunderstand the capabilities of sophisticated adversaries. Practical defense solutions must maintain both machine robustness and human intelligibility—a dual constraint addressed through sophisticated CAPTCHA generation rooted in principles from attacks on Automatic Speech Recognition systems (Abdullah et al., 2023).

This dynamic extends to emerging attack surfaces in large language models. Adversarial training for LLM robustness requires extensive computational resources and task-specific knowledge, limiting practical applicability across diverse deployment scenarios (Shen et al., 2024). More fundamentally, attackers increasingly demonstrate that simple, accessible attacks often prove more

effective than complex gradient-based methods. Dynamic attention mechanisms for Transformer architectures offer more feasible defenses by enhancing robustness without imposing additional computational costs or requiring domain-specific knowledge (Shen et al., 2024).

Privacy-Utility Trade-offs and Scalability Constraints

The persistent tension between privacy protection, model utility, and computational efficiency manifests distinctly across deployment contexts. While Differential Privacy mechanisms like Central Differential Privacy (CDP) and Local Differential Privacy (LDP) demonstrate superior efficacy against both backdoor and membership inference attacks compared to non-private robustness techniques, they impose utility costs that prove unacceptable for small-scale applications or resource-constrained deployments (Naseri et al., 2022). Purely robustness-focused defenses such as gradient norm bounding fail against sophisticated white-box membership inference attacks, revealing that piecemeal security approaches prove inadequate when attackers leverage multiple vulnerability classes simultaneously (Naseri et al., 2022).

Recent work demonstrates that this trade-off is not inevitable through careful mechanism design. HAMP uses high-entropy soft labels and output modification to achieve both high accuracy and strong membership privacy, significantly improving defense feasibility compared to existing approaches that sacrifice substantial utility for privacy guarantees (Chen & Pattabiraman, 2024). However, practical privacy threats extend beyond individual training examples. Real-world privacy auditing often requires determining whether any sample from a particular individual—such as any voice recording of a specific speaker—was used in training, even when the exact samples used for inference differ from those in the training set. This speaker-level membership inference poses a more realistic and challenging threat model than conventional example-level attacks (Chen et al., 2024).

Scalability constraints further complicate practical deployments. When privacy violations are discovered in deployed models, remediation through machine unlearning must often address large groups of features or labels distributed across many data instances. Prior unlearning methods fail to scale efficiently to these realistic remediation scenarios, rendering them practically infeasible despite

their theoretical soundness (Warnecke et al., 2023). Similarly, robust aggregation methods in Federated Learning must ensure fast convergence despite ongoing attacks to meet the high-throughput demands of practical classification tasks, a constraint that traditional robustness-focused aggregation algorithms often violate (Chu et al., 2023).

Adversarial Loops and Counteradaptation

An underappreciated aspect of the deployment gap involves adversarial loops where defenses inadvertently create new attack vectors. Active defenses that modify training processes to detect attacks can themselves become targets for adaptive counterattacks. SplitGuard, designed as an active robust defense against split learning hijacking, was shown to enable a sophisticated counterattack (SplitSpy) that exploits the very mechanisms SplitGuard introduces for protection. This vulnerability loop emphasizes the need for robust, passive inspection techniques that do not tamper with training processes and thus avoid creating exploitable side channels (Fu et al., 2023). Similarly, gradient compression techniques intended to optimize communication efficiency in Federated Learning can inadvertently create vulnerabilities or provide a false sense of security against certain privacy attacks.

Expanding the Adversarial Scope Beyond Classification

Traditional adversarial machine learning research has focused overwhelmingly on discriminative models, particularly image classifiers, leaving other model architectures and task types largely unexplored. This narrow focus creates blind spots in security research as industry deploys increasingly diverse model types. Property inference attacks against Generative Adversarial Networks (GANs) reveal vulnerabilities relevant to intellectual property protection and fairness auditing for generative AI—concerns that classifier-focused security research fails to address (Zhou et al., 2022). As generative models proliferate in commercial applications, understanding their unique vulnerability profiles becomes essential for comprehensive security.

The expansion of adversarial scope extends to multimodal and specialized architectures. Dataset distillation, which compresses large datasets into small

synthetic surrogates, introduces novel attack surfaces where adversaries can poison the distillation process itself, affecting all downstream models trained on the distilled data (Liu et al., 2023). These system-level and architectural vulnerabilities require fundamentally different analytical approaches than those developed for traditional supervised classifiers.

References

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Yearly Breakdown

1.) 2022

The landscape of adversarial machine learning (AML) research shows a clear movement toward addressing practical limitations that hinder the application of robust defenses and the execution of stealthy attacks in real-world systems, moving beyond purely theoretical or digital vulnerabilities. A significant aspect of this research-practice gap lies in transitioning from studying digital adversarial examples (AEs) to ensuring robustness against physical and operational constraints. For instance, conventional studies yielded prolific theoretical results on AEs but often failed to consider the complex physical conditions—such as distance, angle, and illumination changes—that severely diminish the effectiveness of digital AEs when printed and viewed by real systems. To bridge

this gap, research now focuses on creating **robust physical AEs** capable of deceiving safety-critical applications like Traffic Sign Recognition (TSR) systems in autonomous vehicles. A notable effort presents a systematic pipeline to generate robust physical AEs, which proved effective and robust in extensive real-road driving tests against a YOLO v5 based TSR system, successfully fooling a production-grade 2021 model vehicle, thus demonstrating a direct threat to industry technology (Jia et al., 2022).

Beyond physical constraints, the research-industry gap is defined by operational complexities such as distributed training and the need for **stealthy attacks**. In collaborative environments like Federated Learning (FL), adversaries often opt for highly stealthy model poisoning attacks. The newly proposed Model Hijacking attack, for example, prioritizes stealth by leveraging a novel model structure, the Camouflager, to make malicious data samples appear visually similar to benign training data. This ensures the attack is inconspicuous to the model owner while maintaining the utility of the original task, thereby enabling the execution of a completely separate malicious task without detection (Salem et al., 2022).

Furthermore, practical security research is increasingly required to handle the heterogeneity of production environments. Traditional FL defense mechanisms, which often employ outlier detection strategies, fail in real-world scenarios where client data is genuinely heterogeneous (non-IID) and falsely exclude benign models, resulting in utility degradation. Addressing this, DeepSight utilizes model inspection to accurately cluster and identify malicious updates based on homogeneity artifacts, successfully mitigating state-of-the-art backdoor attacks without negatively impacting the performance of the aggregated model for clients with deviating data distributions (Rieger et al., 2022).

This shift toward practical concerns also extends the scope of AML beyond traditional discriminative models and rudimentary security notions. Most prior attacks focused overwhelmingly on classifiers, leaving generative models largely unexplored. Zhou et al. (2022) proposed the first **property inference attacks against Generative Adversarial Networks (GANs)**, focusing on the highly practical black-box setting, thus revealing vulnerabilities relevant to intellectual property protection and fairness auditing for generative AI models. Finally, research evaluating defenses must now operate within complex constraints, acknowledging that real-world deployment requires balancing utility, robustness, and privacy simultaneously. Research shows that purely robustness-focused

defenses, such as Norm Bounding (clipping), often fail against sophisticated inference attacks like white-box membership inference. Consequently, solutions integrating Differential Privacy (DP), such as Central Differential Privacy (CDP) and Local Differential Privacy (LDP), are being investigated as comprehensive defenses, demonstrating better efficacy against both backdoor and membership inference attacks compared to non-private robustness techniques, although facing persistent trade-offs regarding acceptable utility loss, especially in small-scale applications (Naseri et al., 2022).

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2.) 2023

A critical research-to-industry gap persists in adversarial machine learning (AML) due to the mismatch between theoretical constraints in academia and the complex realities of commercial deployment—a theme strongly reflected across NDSS 2023 papers. Conceptually, much academic AML research relies on simplified or inappropriate metrics. For instance, focusing on input imperceptibility using mathematical distances (L_p norms) is often irrelevant for real-world tabular data domains like fraud detection. Here, the true constraints are the quantifiable **financial cost** or **utility** an adversary gains from a successful attack (Kireev et al., 2023). This gap is further exposed by flawed adversarial assumptions: traditional defenses are defeated by evolving threats that violate their core premises, such as defenses assuming a universal, sample-agnostic trigger, which are rendered useless against modern **dynamic backdoor attacks** (Ma et al., 2023). Similarly,

audio defenses fail because they misunderstand the actual capabilities of sophisticated adversaries, relying on simple noise rather than developing practical solutions that maintain both machine robustness and human intelligibility—a challenge addressed by sophisticated CAPTCHA generation rooted in ASR attack principles (Abdullah et al., 2023). Furthermore, the integrity of security research itself has been questioned by studies revealing that foundational data often conflates adversarial behaviors with benign ones (e.g., in browser fingerprinting), leading to inaccurate conclusions about real-world attacks and defense requirements (Wu et al., 2023).

Operationally, industrial architecture and efficiency demands continually outpace current defense methodologies, compounding the research gap. The reliance on legacy security tools integrated solely at the high-level framework stage (e.g., TensorFlow or PyTorch) means defenses often fail once DNNs are compiled into **executables** for deployment on edge devices and cloud services. This necessitates security solutions embedded directly within the deep learning compiler pipeline (Chen et al., 2023). This architectural challenge is acute in the emerging Encoder as a Service (EaaS) model, where clients cannot use rigorous certified robustness techniques due to lacking **white-box access** to the encoder, coupled with the prohibitive **communication overhead** required for verification (Qu et al., 2023). Efficiency constraints also undermine security when large-scale information leaks occur: prior methods for machine unlearning fail to scale efficiently when remediation must address large groups of features or labels spread across many data instances, making the task practically infeasible (Warnecke et al., 2023). Moreover, adversarial research must incorporate external physical and architectural constraints, such as overcoming the inherent denoising effect of standard video compression, which automatically invalidates most adversarial perturbations targeting video recognition in real-world streaming services (Chang et al., 2023). The pressure of practical deployment also highlights vulnerability loops: an actively robust defense against split learning hijacking (SplitGuard) was shown to create an exploit vector for an adaptive counterattack (SplitSpy), emphasizing the need for robust, passive inspection techniques that do not tamper with training processes (Fu et al., 2023). Finally, industrial systems like DL-based traffic analysis present unique constraints—specifically, highly **heterogeneous feature spaces** and the critical need for **real-time** operation—that cannot be met by conventional robustness certification

methods designed around single metrics like L_p norms (Wang et al., 2023). Addressing these vulnerabilities requires leveraging the reality of industrial deployment, such as mitigating privacy leakage arising from the **statistical heterogeneity** inherent in Federated Learning pipelines (Zhou et al., 2023) or using novel robust aggregation methods to ensure fast convergence despite attacks, meeting the high-speed demands of practical classification tasks (Chu et al., 2023). These findings collectively reveal that AML research must move beyond theoretical benchmarks to incorporate domain-specific physics, costs, architectural limitations, and adversarial sophistication to successfully bridge the enduring gap with industry.

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Author(s)	Year	Title/Context
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Chang, J. W., Javaheripi, M., Hidano, S., & Koushanfar, F.	2023	RoVISQ: Reduction of Video Service Quality via Adversarial Attacks on Deep Learning-based Video Compression.
Chen, Y., Yuan, Y., & Wang, S.	2023	OBSAN: An Out-Of-Bound Sanitizer to Harden DNN Executables.
Chu, T., Garcia-Recuero, A., Iordanou, C., Smaragdakis, G., & Laoutaris, N.	2023	Securing Federated Sensitive Topic Classification against Poisoning Attacks.
Fu, J., Ma, X., Zhu, B. B., Hu, P., Zhao, R., Jia, Y., Xu, P., Jin, H., & Zhang, D.	2023	Focusing on Pinocchio's Nose: A Gradients Scrutinizer to Thwart Split-Learning Hijacking Attacks Using Intrinsic Attributes.
Kireev, K., Kulynych, B., & Troncoso, C.	2023	Adversarial Robustness for Tabular Data through Cost and Utility Awareness.
Liu, Y., Li, Z., Backes, M., Shen, Y., & Zhang, Y.	2023	Backdoor Attacks Against Dataset Distillation.

Ma, W., Wang, D., Sun, R., Xue, M., Wen, S., & Xiang, Y.	2023	The "Beatrix" Resurrections: Robust Backdoor Detection via Gram Matrices.
Qu, W., Jia, J., & Gong, N. Z.	2023	REaaS: Enabling Adversarially Robust Downstream Classifiers via Robust Encoder as a Service.
Warnecke, A., Pirch, L., Wressnegger, C., & Rieck, K.	2023	Machine Unlearning of Features and Labels.
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Wu, S., Sun, P., Zhao, Y., & Cao, Y.	2023	Him of Many Faces: Characterizing Billion-scale Adversarial and Benign Browser Fingerprints on Commercial Websites.
Zhou, C., Gao, Y., Fu, A., Chen, K., Dai, Z., Zhang, Z., Xue, M., & Zhang, Y.	2023	PPA: Preference Profiling Attack Against Federated Learning.

15 sources

3.) 2024

Adversarial machine learning (AdvML) research is shifting from theoretical demonstrations—often relying on gradient computation—toward addressing practical challenges in deployed systems and real-world threat models. A primary concern is the high cost and complexity of current defences, particularly for Large Language Models (LLMs). Traditional techniques like adversarial training require extensive computational resources and specific knowledge of downstream tasks, limiting their practical use (Shen et al., 2024). New approaches, such as dynamic attention mechanisms for Transformer architectures, enhance robustness without additional computational or domain knowledge costs, offering more feasible defences for real-world LLM deployment (Shen et al., 2024).

The practical AdvML landscape also demands solutions aligned with real-world attack complexity and privacy concerns. Previous research on Membership

Inference Attacks (MIAs) focused on example-level threats, but actual security auditing often requires identifying if *any* voice sample of a speaker was used in training, even when the specific samples used for inference differ (Chen et al., 2024). Methodologies like speaker-level MIA address this gap, enhancing the practicality of privacy auditing for Speaker Recognition Systems (Chen et al., 2024). Effective defence also requires overcoming the privacy-utility trade-off. One study introduces HAMP, a mechanism using high-entropy soft labels and output modification to achieve both high accuracy and strong membership privacy, significantly improving the feasibility of securing models against inference threats compared to existing defences (Chen et al., 2024).

The gap persists in the inadequacy of current detection tools against modern threats and the reliance on simplified adversarial models for defence evaluation. The rapid adoption of technologies like ChatGPT has created urgent demand for detection tools to address societal concerns like plagiarism and misinformation—challenges where existing detectors often lack precision and generality (Kumari et al., 2024). Furthermore, evaluation methodologies are often unrealistic. Conventional academic models of adaptive adversaries rely on manually tuned parameters and fail to account for complex, multi-constraint threats encountered in reality (Krauβ et al., 2024). To counter this, automated and comprehensive evaluation frameworks like AutoAdapt are necessary to establish rigorous baselines for testing robustness against adaptive attackers operating under realistic constraints, driving research toward defences that truly withstand sophisticated real-world strategies (Krauβ et al., 2024).

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Is that all the references?

The references provided in the previous response—Shen et al. (2024), Kumari et al. (2024), Krauß et al. (2024), Chen et al. (2024), and Chen & Pattabiraman (2024)—were limited to those papers explicitly summarized in the brief literature review. The collected source material, however, contains excerpts from many more papers presented at the NDSS 2024 Symposium.